

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	-
Project Name	Smart lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Dataset Loading	Load the loan prediction dataset using Pandas	Data Collection	Done	High
2	Data Preprocessing	Handle missing values, encoding, scaling, and balancing	Data Preprocessing	Done	High
3	Data Visualization	Perform univariate, bivariate, and multivariate analysis	Data Analysis	Done	Medium
4	Decision Tree Model	Train and evaluate the Decision Tree classifier	Machine Learning	Done	High
5	Random Forest Model	Train and evaluate the Random Forest classifier	Machine Learning	Done	High
6	KNN Model	Train and test the K-Nearest Neighbors model	Machine Learning	Done	High
7	Gradient Boosting Model	Train and compare the Gradient Boosting model	Machine Learning	Done	High
8	Loan Prediction Web Application	Build a Flask web application with HTML pages for prediction	Flask Application	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Page
2	Backend / Logic	4	Flask Routing, Prediction Logic, Model Loading, Input Validation
3	Database / Storage	1	Dataset (.csv) and Trained Model (.pkl)
4	API / Integration	1	Flask Prediction API (/predict)
5	Security / Authentication	1	Input Validation and Error Handling